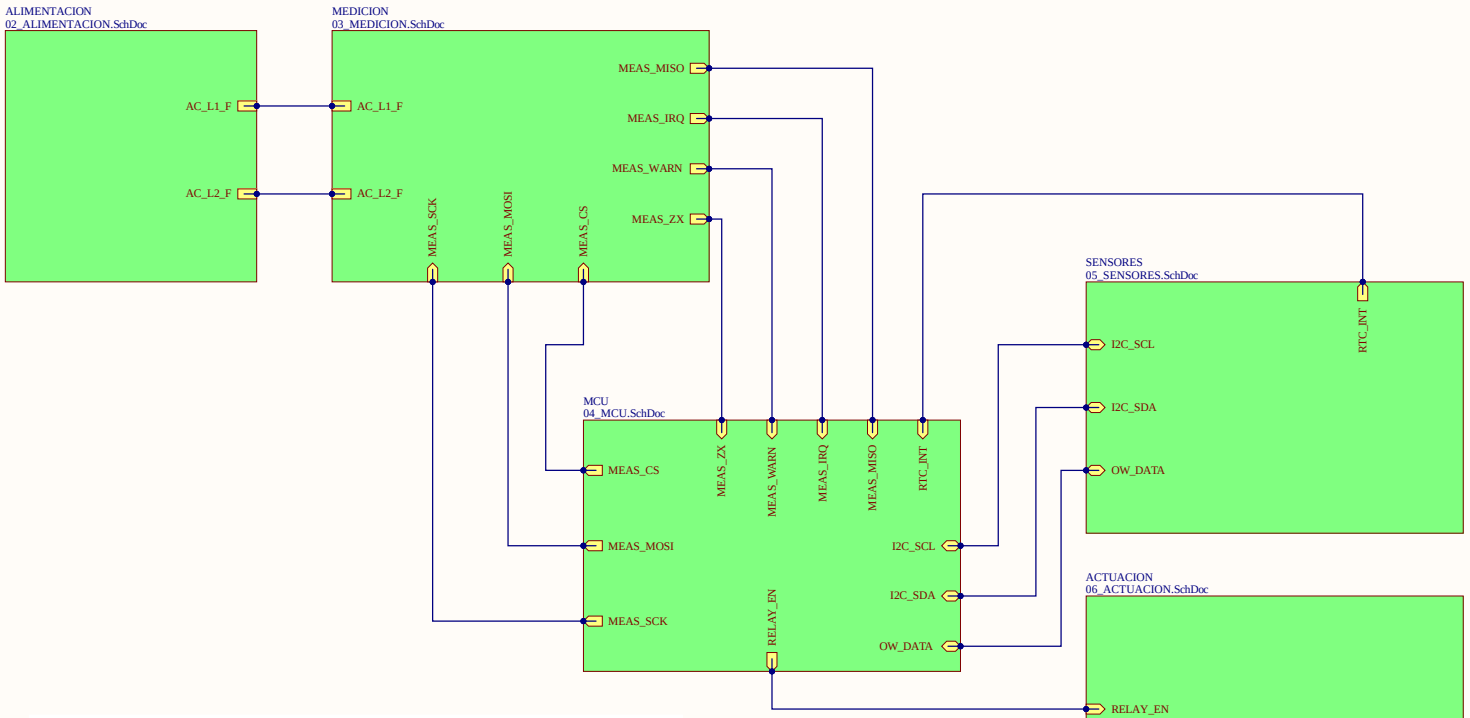
CONTROL DE CARGA EVSE · HARDWARE EN IOT 2026
DISEÑO: NICOLÁS GÓMEZ RAMÍREZ
ZONA DE RED 120–240 VAC CON AISLAMIENTO REFORZADO
USO ACADÉMICO, SIN GARANTÍA.

REVISION	DESCRIPTION	DATE	APPROVED
1.0	Diagrama de bloques (hoja TOP)	10/19/2026	

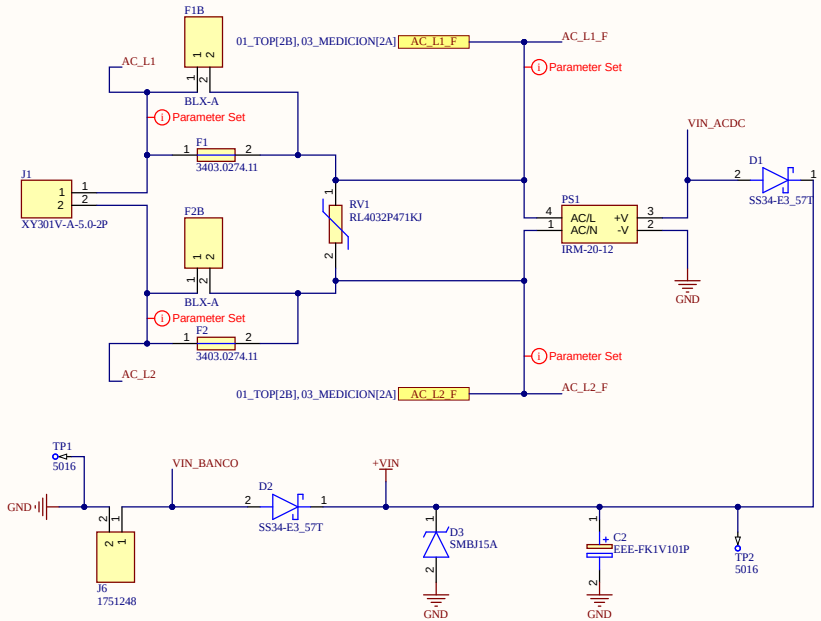


Rieles (puertos de alimentación globales, sin entradas de hoja):
+VIN (12 V): ALIMENTACION → ACTUACION
+3V3: ALIMENTACION → MEDICION, MCU, SENSORES
+3V3A: ALIMENTACION → MEDICION · GND: todas las hojas SELV
AC_L1_F / AC_L2_F = red 120–240 VAC (clase HV_AC). En 120 V, AC_L2_F es el neutro.
Clases de red: HV_AC (red), ANALOG (medición), EVSE_EXT (J2 al EVSE)

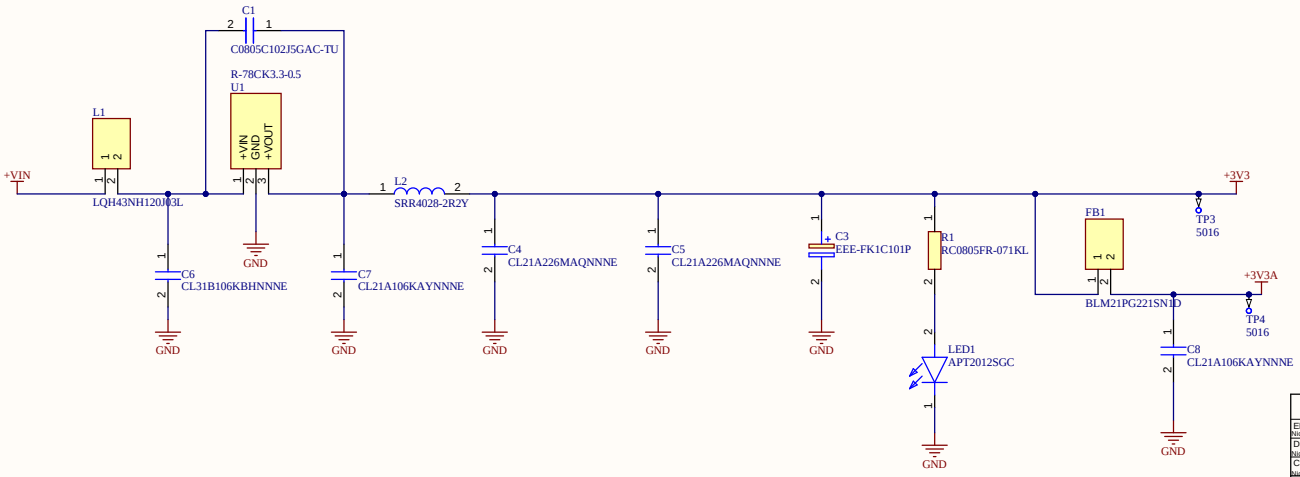
APPROVALS	DATE	PROJECT	Hardware en IoT · 2026
ENG: Nicolás Gómez Ramírez	10/19/2026	Control de Carga IoT.PrjPcb	
DSN: Nicolás Gómez Ramírez	10/19/2026	PROJECT REVISION: 1.0	DOCUMENT REVISION: Not in version control
CHK: Nicolás Gómez Ramírez	10/19/2026	DESIGN BY: Nicolás Gómez Ramírez	
REFERENCE DOCUMENTS		TITLE	Diagrama de bloques (hoja TOP)
BOM:			
ASSY DWG:		SIZE A3	CAGE CODE / DWG NO. / REV
FAB DWG:			
PCB DWG:		SCALE:	FILE NAME 01_TOP.SchDoc
		SHEET 1 OF 6	

CONTROL DE CARGA EVSE · HARDWARE EN IOT 2026
DISEÑO: NICOLÁS GÓMEZ RAMÍREZ
ZONA DE RED 120–240 VAC CON AISLAMIENTO REFORZADO
USO ACADÉMICO, SIN GARANTÍA.

REVISION	DESCRIPTION	DATE	APPROVED
1.0	Alimentación: red, AC-DC 12 V y 3,3 V	10/19/2026	



Alimentar desde breaker exclusivo de 2–6 A (bipolar en 240 V). J1: 1 = L1/L, 2 = L2/N. J6: 12 V de banco (9–15 V).

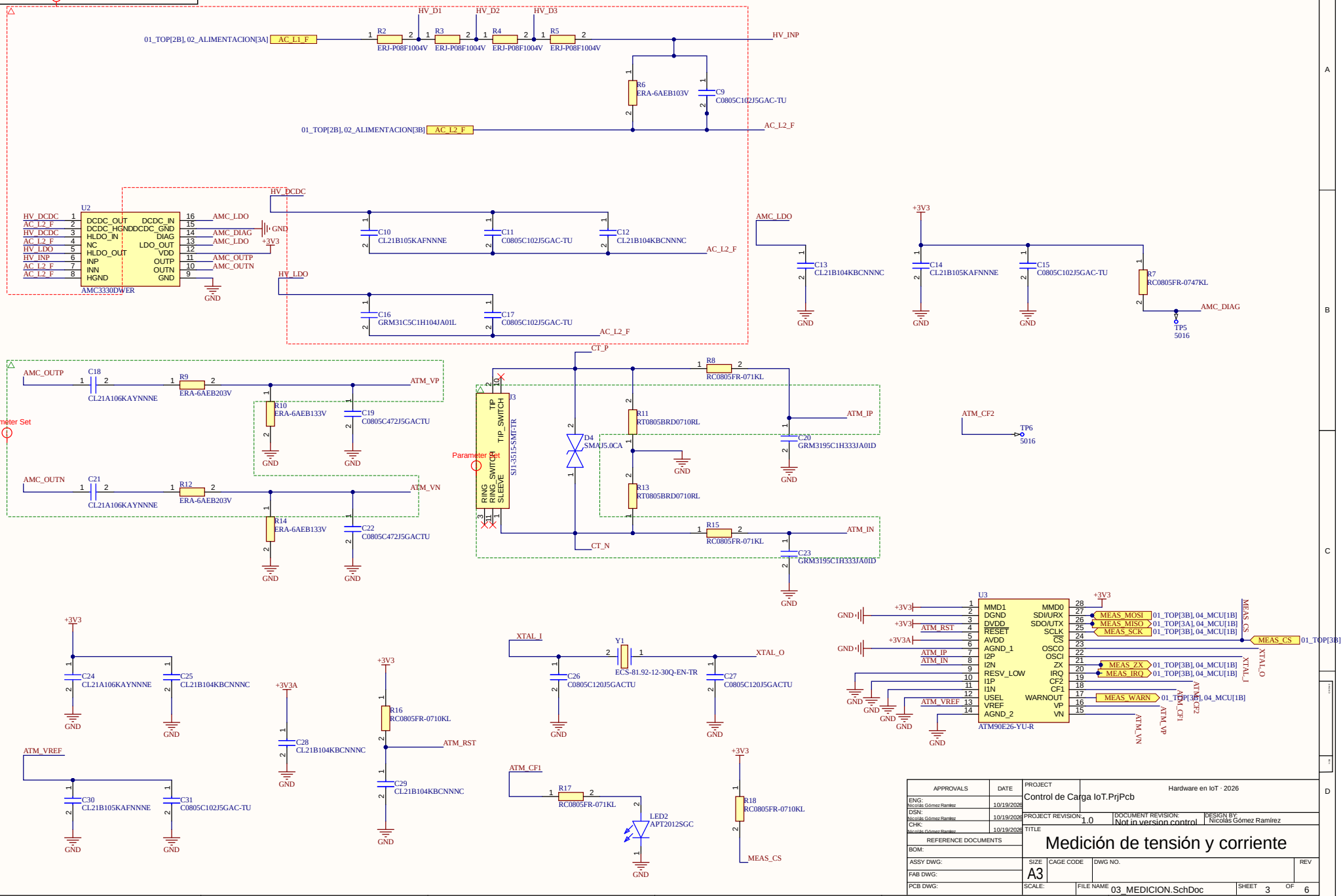


APPROVALS	DATE
ENG: Nicolás Gómez Ramírez	10/19/2026
DSN: Nicolás Gómez Ramírez	10/19/2026
CHK: Nicolás Gómez Ramírez	10/19/2026
REFERENCE DOCUMENTS	
BOM:	
ASSY DWG:	
FAB DWG:	
PCB DWG:	

PROJECT: Control de Carga IoT.PrjPcb — Hardware en IoT · 2026
PROJECT REVISION: 1.0 — DOCUMENT REVISION: Not in version control — DESIGN BY: Nicolás Gómez Ramírez
TITLE: Alimentación: red, AC-DC 12 V y 3,3 V
SIZE: A3 — CAGE CODE — DWG NO. — REV
SCALE: — FILE NAME: 02_ALIMENTACION.SchDoc — SHEET 2 OF 6

CONTROL DE CARGA EVSE · HARDWARE EN IOT 2026
DISEÑO: NICOLÁS GÓMEZ RAMÍREZ
ZONA DE RED 1 (BAJA TENSIÓN) CON AISLAMIENTO REFORZADO
USO ACADÉMICO, SIN GARANTÍA.

REVISION	DESCRIPTION	DATE	APPROVED
1.0	Medición de tensión y corriente	10/19/2026	



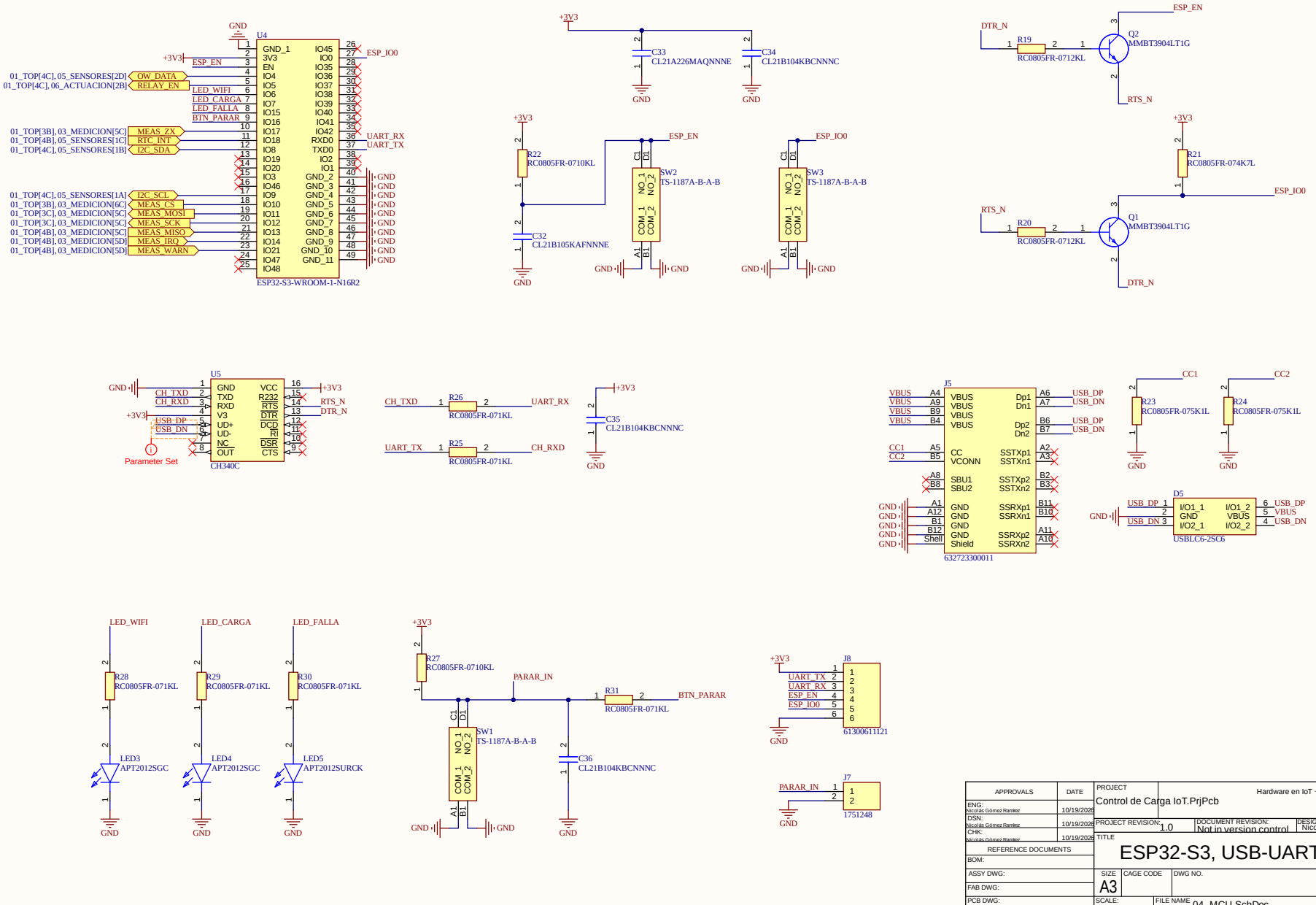
APPROVALS	DATE
ENG: Nicolás Gómez Ramírez	10/19/2026
DSN: Nicolás Gómez Ramírez	10/19/2026
CHK: Nicolás Gómez Ramírez	10/19/2026

REFERENCE DOCUMENTS
BOM:
ASSY DWG:
FAB DWG:
PCB DWG:

PROJECT: Control de Carga IoT.PrjPcb
Hardware en IoT · 2026
PROJECT REVISION: 1.0
DOCUMENT REVISION: Not in version control
DESIGN BY: Nicolás Gómez Ramírez
TITLE: Medición de tensión y corriente
SIZE: A3
CAGE CODE
DWG NO.
REV
SCALE:
FILE NAME: 03_MEDICION.SchDoc
SHEET 3 OF 6

CONTROL DE CARGA EVSE · HARDWARE EN IOT 2026
DISEÑO: NICOLÁS GÓMEZ RAMÍREZ
ZONA DE RED 120–240 VAC CON AISLAMIENTO REFORZADO
USO ACADÉMICO, SIN GARANTÍA.

REVISION	DESCRIPTION	DATE	APPROVED
1.0	ESP32-S3, USB-UART y programación	10/19/2026	



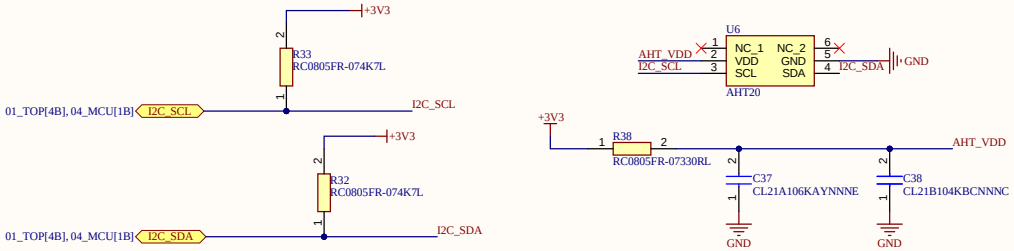
APPROVALS	DATE
ENG: Nicolás Gómez Ramírez	10/19/2026
DSN: Nicolás Gómez Ramírez	10/19/2026
CHK: Nicolás Gómez Ramírez	10/19/2026

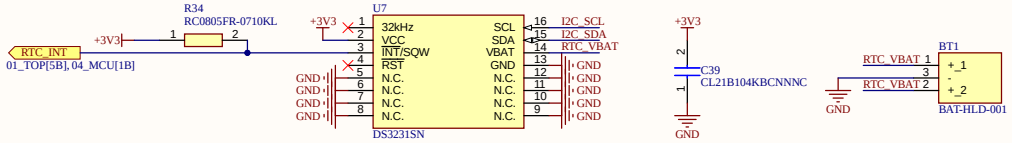
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BOM:	
ASSY DWG:	
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PCB DWG:	

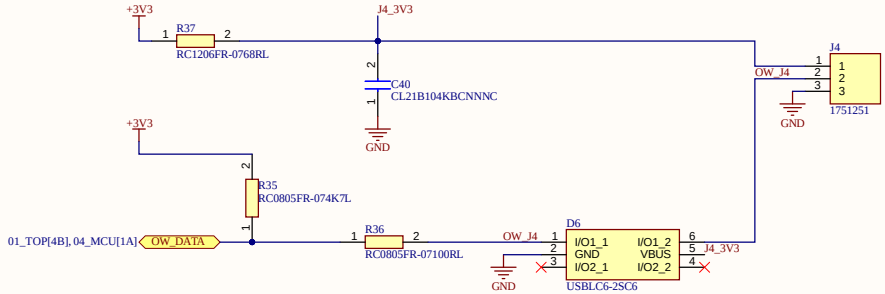
PROJECT: Control de Carga IoT.PrjPcb — Hardware en IoT · 2026
PROJECT REVISION: 1.0
DOCUMENT REVISION: Not in version control
DESIGN BY: Nicolás Gómez Ramírez
TITLE: ESP32-S3, USB-UART y programación
SIZE: A3
CAGE CODE:
DWG NO.:
REV:
SCALE:
FILE NAME: 04_MCU.SchDoc
SHEET 4 OF 6

CONTROL DE CARGA EVSE · HARDWARE EN IOT 2026
DISEÑO: NICOLÁS GÓMEZ RAMÍREZ
ZONA DE RED 120–240 VAC CON AISLAMIENTO REFORZADO
USO ACADÉMICO, SIN GARANTÍA.

REVISION	DESCRIPTION	DATE	APPROVED
1.0	Sensores, RTC y sonda 1-Wire	10/19/2026	





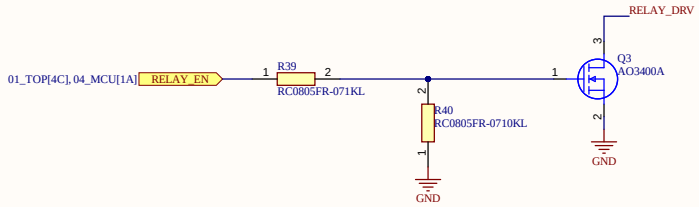


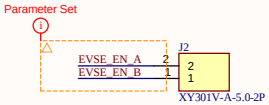
APPROVALS	DATE
ENG: Nicolás Gómez Ramírez	10/19/2026
DSN: Nicolás Gómez Ramírez	10/19/2026
CHK: Nicolás Gómez Ramírez	10/19/2026
REFERENCE DOCUMENTS	
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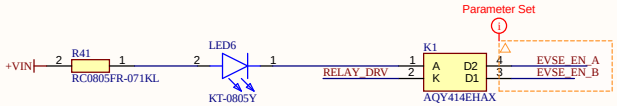
PROJECT: Control de Carga IoT.PrjPcb — Hardware en IoT · 2026
PROJECT REVISION: 1.0 — DOCUMENT REVISION: Not in version control — DESIGN BY: Nicolás Gómez Ramírez
TITLE: Sensores, RTC y sonda 1-Wire
SIZE: A3 — CAGE CODE — DWG NO. — REV
SCALE: — FILE NAME: 05_SENSORES.SchDoc — SHEET 5 OF 6

CONTROL DE CARGA EVSE · HARDWARE EN IOT 2026
DISEÑO: NICOLÁS GÓMEZ RAMÍREZ
ZONA DE RED 120–240 VAC CON AISLAMIENTO REFORZADO
USO ACADÉMICO, SIN GARANTÍA.

REVISION	DESCRIPTION	DATE	APPROVED
1.0	Actuación: PhotoMOS K1 hacia el EVSE	10/19/2026	







APPROVALS	DATE
ENG: Nicolás Gómez Ramírez	10/19/2026
DSN: Nicolás Gómez Ramírez	10/19/2026
CHK: Nicolás Gómez Ramírez	10/19/2026
REFERENCE DOCUMENTS	
BOM:	
ASSY DWG:	
FAB DWG:	
PCB DWG:	

PROJECT: Control de Carga IoT.PrjPcb — Hardware en IoT · 2026
PROJECT REVISION: 1.0 — DOCUMENT REVISION: Not in version control — DESIGN BY: Nicolás Gómez Ramírez
TITLE: Actuación: PhotoMOS K1 hacia el EVSE
SIZE: A3 — CAGE CODE — DWG NO. — REV
SCALE: — FILE NAME: 06_ACTUACION.SchDoc — SHEET 6 OF 6

Comment	Description	Designator	Footprint	LibRef	Quantity
BAT-HLD-001	Battery holder CR2032, SMD	BT1	BATHLD001	BAT-HLD-001	1
C0805C102JGAC-TU	Capacitor	C1, C9, C11, C15, C17, C31	CAPC2012X88N	C0805C102JGAC-TU	6
EEE-FK1V101P	Capacitor Polarised	C2	EEHZF1E331UP	EEE-FK1V101P	1
EEE-FK1C101P	Capacitor Polarised	C3	EEEFLA1D151P	EEE-FK1C101P	1
CL21A226MAQNNNE	Capacitor	C4, C5, C33	CAPC2012X140N	CL21A226MAQNNNE	3
CL31B106KBHNNNE	Capacitor	C6	CAPC3216X180N	CL31B106KBHNNNE	1
CL21A106KAYNNNE	Capacitor	C7, C8, C18, C21, C24, C37	CAPC2012X145N	CL21A106KAYNNNE	6
CL21B105KAFNNNE	Capacitor	C10, C14, C30, C32	CAPC2012X88N	CL21B105KAFNNNE	4
CL21B104KBCNNNC	Capacitor	C12, C13, C25, C28, C29, C34, C35, C36, C38, C39, C40	CAPC2012X88N	CL21B104KBCNNNC	11
GRM31C5C1H104JA01L	Capacitor ceramico 100 nF 50 V C0G 5% 1206	C16	CAPC3216X180N	GRM31C5C1H104JA01L	1
C0805C472JGACTU	Capacitor	C19, C22	CAPC2012X88N	C0805C472JGACTU	2
GRM3195C1H333JA01D	Capacitor ceramico 33 nF 50 V C0G 5% 1206	C20, C23	CAPC3216X180N	GRM3195C1H333JA01D	2
C0805C120JGACTU	Capacitor	C26, C27	CAPC2012X88N	C0805C120JGACTU	2
SS34-E3_57T	Schottky Diode	D1, D2	DIOM7959X262N	SS34-E3_57T	2
SMBJ15A	TVS Diode (Uni-directional)	D3	DIOM5436X244N	SMBJ15A	1
SMAJ5.0CA	TVS Diode (Bi-directional)	D4	DIONM5026X229N	SMAJ5.0CA	1
USBLC6-2SC6	Integrated Circuit	D5, D6	SOT95P280X145-6N	USBLC6-2SC6	2
3403.0274.11	Fusible SMD UMT-H T1A 277 VAC, ruptura 1500 A (bolsa)	F1, F2	3403027423	3403.0274.11	2
BLX-A	Portafusible 5x20 mm para PCB, 2 pines a 22 mm (XC Electronics BLX-A, LCSC C3131)	F1B, F2B	BLX-A_5X20	BLX-A	2
BLM21PG221SN1D	Ferrite Bead	FB1	BEADC2012X105N	BLM21PG221SN1D	1
XY301V-A-5.0-2P	Bornera de tornillo 2 polos, paso 5.0 mm, 300 V 15 A, 22-14 AWG	J1, J2	TB001-500-02BE	XY301V-A-5.0-2P	2
SJ1-3515-SMT-TR	Connector	J3	SJ13515SMTTR	SJ1-3515-SMT-TR	1
1751251	Connector	J4	1751251	1751251	1
632723300011	Conn USB 3.1 Type C RCP 24 POS 0.5mm/0.4mm Solder R/A SMD/Thru-Hole 24 Terminal 1 Port T/R	J5	632723300011	CMP-1502-03214-4	1
1751248	Connector	J6, J7	1751248	1751248	2
61300611121	Connector	J8	HDRMBW79P0X254_1X6_1524X254X894P	61300611121	1
AQY414EHAX	PhotoMOS 1 Form B (NC), 400 V 120 mA, reinforced insulation 5000 Vrms, SMD DIP-4	K1	SOP254P962X315-4N	AQY-414EHAX	1
LQH43NH120J03L	Inductor 12 uH 5% 420 mA 0.25 ohm 1812, AEC-Q200	L1	RLS126	LQH43NH120J03L	1
SRN4028-2R2Y	Inductor	L2	SRN4028	SRN4028-2R2Y	1
APT2012SGC	LED	LED1, LED2, LED3, LED4	LEDC2012X85N	APT2012SGC	4
APT2012SURCK	LED	LED5	LEDC2012X85N	APT2012SURCK	1
KT-0805Y	LED yellow 0805, 590 nm	LED6	LEDC2012X85N	KT-0805Y	1
IRM-20-12	Power Supply	PS1	IRM2012	IRM-20-12	1
MMBT3904LT1G	Transistor BJT NPN	Q1, Q2	SOT96P237X111-3N	MMBT3904LT1G	2
AO3400A	MOSFET N-channel 30V 5.7A, Rds(on) 48mR max at Vgs 2.5V, SOT-23	Q3	SOT96P237X111-3N	AO3400A	1
RC0805FR-071KL	Resistor	R1, R8, R15, R17, R25, R26, R28, R29, R30, R31, R39, R41	RESC2012X60N	RC0805FR-071KL	12
ERJ-P08F1004V	Resistor	R2, R3, R4, R5	ERJP08_(1206)	ERJ-P08F1004V	4
ERA-6AEB103V	Resistor	R6	RESC2012X60N	ERA-6AEB103V	1
RC0805FR-0747KL	Resistor	R7	RESC2012X60N	RC0805FR-0747KL	1
ERA-6AEB203V	Resistor	R9, R12	RESC2012X60N	ERA-6AEB203V	2
ERA-6AEB133V	Resistor	R10, R14	RESC2012X60N	ERA-6AEB133V	2
RT0805BRD0710RL	Resistencia de película delgada 10 ohm 0.1% 25 ppm 0805	R11, R13	RESC2012X60N	RT0805BRD0710RL	2
RC0805FR-0710KL	Resistor	R16, R18, R22, R27, R34, R40	RESC2012X60N	RC0805FR-0710KL	6
RC0805FR-0712KL	Resistor	R19, R20	RESC2012X60N	RC0805FR-0712KL	2
RC0805FR-074K7L	Resistor	R21, R32, R33, R35	RESC2012X60N	RC0805FR-074K7L	4
RC0805FR-075K1L	Resistor	R23, R24	RESC2012X60N	RC0805FR-075K1L	2
RC0805FR-07100RL	Resistor	R36	RESC2012X60N	RC0805FR-07100RL	1
RC1206FR-0768RL	Resistor	R37	ERJP08_(1206)	RC1206FR-0768RL	1
RC0805FR-07330RL	Resistor	R38	RESC2012X60N	RC0805FR-07330RL	1
RL4032P471KJ	Varistor SMD 300 VAC, 470 V, 2500 A (8/20 us), 30 J, 4032	RV1	CL4032	RL4032P471KJ	1
TS-1187A-B-A-B	Tact switch 5.1 x 5.1 x 1.5 mm, 160 gf, 50 mA 12 VDC	SW1, SW2, SW3	TS1187ABAB	TS-1187A-B-A-B	3
5016	Test Point, 1-Pin SMT, RoHS, Tape and Reel	TP1, TP2, TP3, TP4, TP5, TP6	KEYSTONE-5016_V	CMP-2000-05178-1	6
R-78OK3.3-0.5	Regulador conmutado 3.3 V 0.5 A, SIP3	U1	R78OK3305	R-78OK3.3-0.5	1
AMC3330DWER	Amplificador aislado +-1 V con DC/DC integrado, aislamiento reforzado, SOIC-16 DWE(carrete)	U2	SOIC127P1030X265-16N	AMC3330DWER	1
ATM90E26-YU-R	Integrated Circuit	U3	SOP65P778X199-28N	ATM90E26-YU-R	1
ESP32-S3-WROOM-1-N16R2	Integrated Circuit	U4	ESP32S3WROOM1N16R2	ESP32-S3-WROOM-1-N16R2	1
CH340C	USB to UART bridge	U5	SOIC127P600X175-16N	CH340C	1
AHT20	Integrated Circuit	U6	AHT20	AHT20	1
DS3231SN	I2C RTC with integrated TCXO and crystal, VBAT, SO-16W	U7	SOIC127P1030X265-16N	DS3231SN	1
ECS-81.92-12-30Q-EN-TR	Crystal or Oscillator	Y1	ECS81921230QENTR	ECS-81.92-12-30Q-EN-TR	1